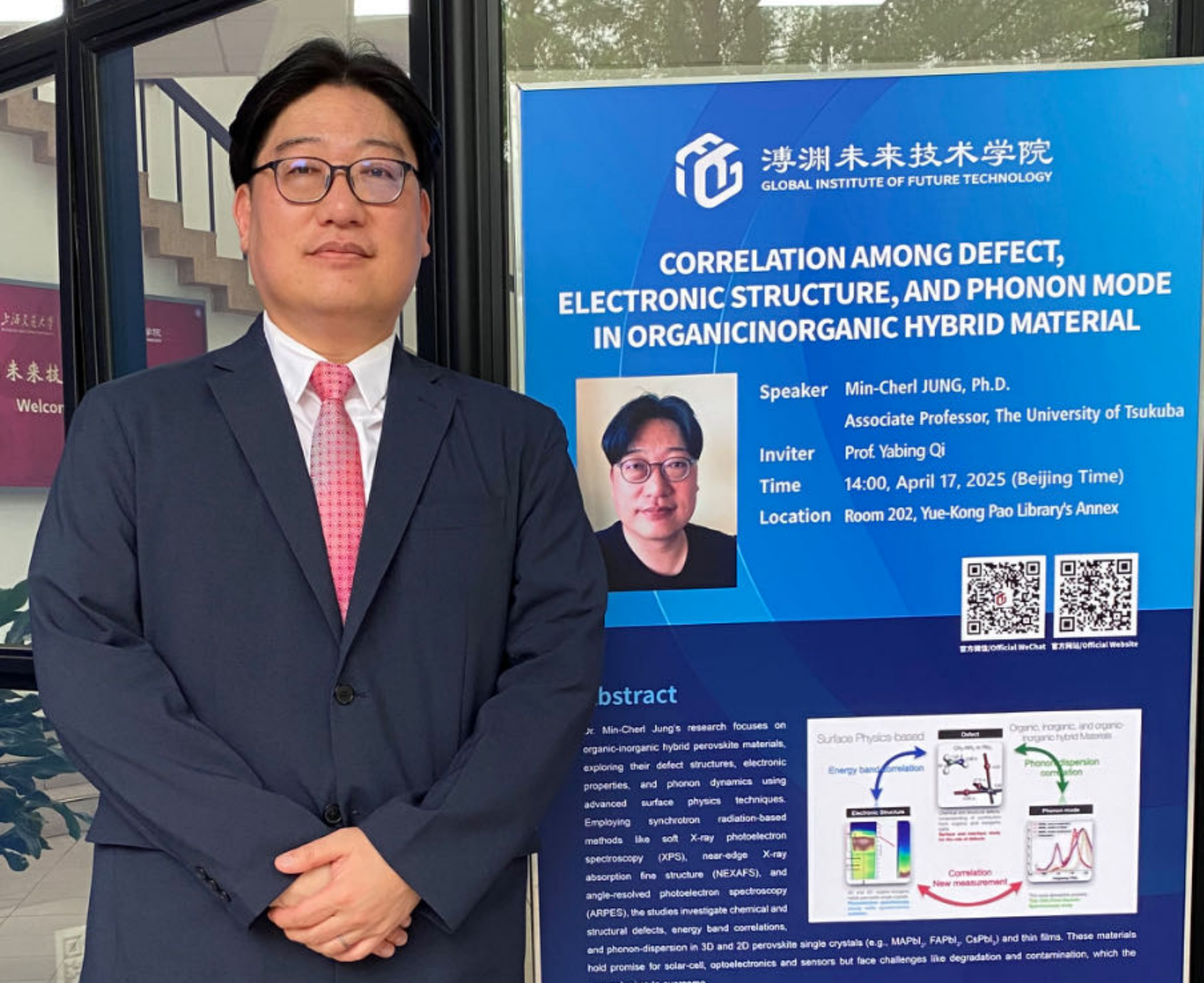
溥渊未来技术学院
GLOBAL INSTITUTE OF FUTURE TECHNOLOGY
CORRELATION AMONG DEFECT, ELECTRONIC STRUCTURE, AND PHONON MODE IN ORGANICINORGANIC HYBRID MATERIAL
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Inviter Prof. Yabing Qi
Time 14:00, April 17, 2025 (Beijing Time)
Location Room 202, Yue-Kong Pao Library's Annex
Official WeChat
Official Website
bstract
Jr. Min-Cherl Jung's research focuses on organic-inorganic hybrid perovskite materials, exploring their defect structures, electronic properties, and phonon dynamics using advanced surface physics techniques. Employing synchrotron radiation-based methods like soft X-ray photoelectron spectroscopy (XPS), near-edge X-ray absorption fine structure (NEXAFS), and angle-resolved photoelectron spectroscopy (ARPES), the studies investigate chemical and structural defects, energy band correlations, and phonon-dispersion in 3D and 2D perovskite single crystals (e.g., MAPbI3, FAPbI3, CsPbI3) and thin films. These materials hold promise for solar-cell, optoelectronics and sensors but face challenges like degradation and contamination, which the
Surface Physics based
Defect
Organic, inorganic, and organic-inorganic hybrid Materials
Energy band correlation
Phonon dispersion correlation
Electronic Structure
Phonon mode
Correlation
New measurement